

JULIA KACZMAREK

Your Next Best Tech Hire

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GitHub | **Linkedin**

SKILLS

Programming Languages:	Python, Bash, HTML, CSS, JavaScript
Libraries/Frameworks:	Qt (PySide6), Pandas, Flask
Software Development:	Desktop Application Development, Object-Oriented Programming, MVC Architecture, Version Control
Tools / Platforms:	Linux, Git, GitHub, LaTeX, Orca (Quantum Chemistry Software), HPC
Databases:	SQL

EDUCATION

The University of Manchester	Manchester, UK
Computational Chemistry PhD (Doctor of Philosophy)	September 2024 - February 2028 (Expected)
Newcastle University	Newcastle-upon-Tyne, UK
Chemistry MChem (Master of Chemistry)	September 2021 - July 2024
Percentage: 75.7%	

EXPERIENCE

The University of Manchester | Academic Researcher Manchester, UK | September 2024 - Present

- Leading development of AQMEasy (PySide6), delivering an application that automates computational chemistry workflows and reduces technical barriers for researchers. This requires a consistently high level of initiative, resilience and problem solving.
- Provided computational chemistry expertise to support experimental reactivity projects across organic and inorganic materials groups. Bridged computational predictions with experimental validation across multiple collaborations requiring high organisation, time management, teamwork and communication skills.
- Presented research at international conferences, translating complex technical findings for diverse audiences and strengthening networking skills.
- Supervised summer student in developing an automation interface for quantum chemistry input files, delivering a functional tool. Continued to mentor the student since by checking in monthly on university and career progress.

The University of Manchester | Graduate Teaching Assistant Manchester, UK | January 2025 - Present

- Supervised weekly computational lab sessions of 20-30 students, teaching Gaussian software and providing technical guidance for computational chemistry calculations.

Newcastle University | Computational Research Intern Newcastle-upon-Tyne, UK | June 2023 - September 2023

- Optimised an LSTM model to predict spectroscopic line-shapes from molecular geometry, achieving an improved accuracy over existing methods of 96%(2% increase). Presented work at local (poster) and national (talk) conferences, with results published in a review paper.
- Gained proficiency in Linux, bash scripting, and machine learning fundamentals.

PROJECTS / OPEN-SOURCE

AQMEasy | Link *Python, Qt*

- Developing a modular desktop GUI for the AQME computational chemistry software, designed to make quantum chemistry calculations accessible to researchers with limited programming experience.
- Automates workflow setup, job submission, and results analysis for common computational chemistry tasks, reducing setup time and technical barriers.

QTAIM Visualiser | Link *Python, py3dmol*

- Developed a Jupyter Notebook tool for visualising non-covalent interactions from QTAIM analysis performed with Multiwfn software.

- Automates atom assignment between interactions and enables parameter-based searching, streamlining analysis of molecular interactions.

NBO Visualiser | [Link](#)

Python, py3dmol

- Developed a Jupyter Notebook tool for visualising Natural Bond Orbital (NBO) analysis results from ORCA+NBO7 calculations, including SOPT and NPA data.
- Interfaces with QTAIM Visualiser to compare and validate non-covalent interaction predictions across different analysis methods, saving multiple hours of work.

twordle (Terminal Wordle) | [Link](#)

Python, curses

- Developed a terminal-based implementation of the popular word game Wordle using the curses library for an interactive CLI interface.

CERTIFICATIONS

- Introduction to Data and SQL (Highly Commended Candidate) - **CodeFirstGirls 2023**
- Introduction to Python and Apps - **CodeFirstGirls 2023**

HONORS & AWARDS

- 2024 Neil Hughes Prize at Newcastle University for contributions to the school
- 2023 Levin Prize at Newcastle University for meritorious performance in Stage 3
- 2022 SCI Prize at Newcastle University for meritorious performance in Stage 2